

CHANDANA PAMIDI

+1(947) 229-8633 ◇ pamidichandana81@gmail.com ◇ [Linkedin](#) ◇ [Github profile](#) ◇ Amherst,MA,US

EDUCATION

Masters in Computer Science, University of Massachusetts Amherst, **CGP: 3.529** **Feb 2023 - Dec 2024**

- **Relevant Coursework:** Introduction to Computer and Network Security, Advance Natural Language Processing, Information Retrieval, Intelligent Visual Computing, Machine Learning, Theory and practice of Software Engineering, Meth Applied Stats, Distributed Operating Systems, Advanced Algorithms
- **Graduate Teaching Assistant**, working with prof. Juan Zhai as Grader for CS520: Theory and practice of Software engineering

Bachelors in Computer Science, Lakireddy Balireddy College of Engineering, affiliated to Jawaharlal Nehru Technological University Kakinada, **CGP: 8.92** **Jul 2018 - May 2022**

- **Relevant Coursework:** Information Security, Data Communications and Computer Networks, Data Mining and Data Warehousing, Artificial Intelligence, Image Processing, Software Engineering, Design Patterns, Big Data Analytics, Cloud Computing, Automata Theory and Compiler Design, Design and Analysis of Algorithms, Object Oriented Analysis and Design, Data Base Management Systems, Data Structures
- Secured 2nd place among 300+ participants in the coding contest organized by Tecda club of IT department.

TECHNICAL SKILLS

Programming languages: C, Python, Java, R

Big Data Tools: Hadoop, Spark, Kafka, Hive, Cassandra, Tableau

Operating Systems: Unix/Linux

Web Development: HTML, CSS, Javascript, PHP

Libraries: Pandas, Numpy, TensorFlow, matplotlib, scikit-learn, PyTorch,

Tools: Git, AutoCAD, VS Code, Google colab, Slack, Code blocks, PyCharm,

Cloud Computing: AWS

Query Languages: SQL, GQL

Data Structures and Algorithms

Servers: Apache, XAMPP

OpenCV

IDLE, Jupyter, Anaconda, Eclipse

WORK EXPERIENCE

Advanced App Engineering Associate, Accenture

Aug 2022 - Jan 2023 , Bangalore, IND

- Designed a POC to format 50+ attributes of data corresponding to the accounting details of the client (bank).
- Processed 100TB of ORC data and converted it into Parquet format using Apache PySpark.

Programmer Analyst Intern, Cognizant

Mar 2022 - May 2022 , Bangalore, IND

- Dedicated an average of 40 hours per week in learning about four new technologies, including Python libraries for data visualization, Apache Spark for big data processing, and various AWS services like EC2 and S3.

Summer Intern, Swecha

Jun 2020 - Aug 2020 , Remote

- Developed a study tool benefiting government school students, resulting in zero-cost access to learning materials.
- Used Bootstrap, HTML5, javascript, MySQL to increase students interaction with the tool about 40%.

PROJECTS

Chain-of-thought for complex reasoning using LLM. [link](#)

Feb 2024 - May 2024

- Achieved significant performance improvements by fine-tuning the Codellama-7B model using PEFT with LoRA, resulting in notable gains in accuracy, F1-Score, and ROUGE metrics.
- Implemented custom weighted loss during fine-tuning, resulting in superior model performance across accuracy, F1-Score, ROUGE, and BLEU metrics.
- Delivered robust model comparisons by evaluating under zero-shot, few-shot, and with/without rationale settings, driving data-driven insights for optimized model performance.

Rocky Montana: Analyzing Traffic Violations in the State of Montana. [link](#)

Feb 2024 - May 2024

- Analyzed traffic stops in Montana by performing statistical tests like Chi-Square, linear regression, ANOVA to uncover meaningful insights on 9 years of data using R Studio.

Context-Enhanced Report Generation Using multi-model LLM. [link](#)

May 2024 - May 2024

- Fine-tuned the BLIP T5 model from the LAVIS library on a dataset of 89,400 instances, using prompt-based tuning to improve medical report generation accuracy.
- Deployed the model for real-time inference via a Flask API, enabling seamless integration and interaction with web applications for generating medical reports.

Personalized news categorization using large language model. **Sep 2023 - Dec 2023**

- Refined existing LaMP models with personalization through retrieval augmentation to get accuracy of 58.9%

Performance evaluation of various Machine Learning Algorithms. **Feb 2023 - May 2023**

- Implemented and fine-tuned various machine learning models like Random Forest, Naive Bayes, Neural Networks, etc, with rigorous experiments for optimizing hyperparameters to increase accuracy on an average of 6%.

Elevation based Navigation Aviation (EleNA). **Feb 2023 - May 2023**

- Developed a navigation system adhering to MVC architecture and software design principles, enabled route optimization for elevation gain while limiting total distance traveled to a specified x% of the shortest path.

- Implemented 10+ APIs and tested the code using JUnit which resulted in 90%+ code coverage.

Distributed Systems Development: From Concurrency to Cloud Deployment. **Feb 2023 - May 2023**

- Designed a distributed client-server systems with concurrent networking, employing both socket communication and gRPC for remote procedure calls.

- Implemented REST API servers, effectively containerized micro-services using Docker, and tested distributed systems for reliability, caching, replication, fault tolerance, high availability, and cloud deployment.

- Deployed the microservice in AWS EC2 instance of type m5a.large to increase the latency to 400milliseconds.

BERT for language processing to detect trolls. **Feb 2022 - May 2022**

- Employed BERT to learn contextual relations of the text which resulted in 7.2% increase of detection in trolls.

Face Recognition and Identification. **Feb 2021 - May 2021**

- Implemented Haar Cascade classifier to detect facial features to speeding up the computation by 5 times.

Code Review System. **Jul 2019 - Nov 2019**

- Developed a user interface using HTML, CSS, Javascript, Bootstrap with user authentication and authorization.

- Designed a SQL database schema for efficient data storage and retrieval.

LEADERSHIP

- Managed daily activities and assemblies as School People Leader and led 500+ students.
- Led the school's Rotary Club as President in 2014-15, actively coordinating social welfare activities and events.
- Tackled challenges affecting women within the college community as the Class Representative of Saheli club.